

MIT Assignment Lab

Question 1

Use the matrix method to determine whether the following system of equations has a unique solution, no solution, or infinitely many solutions:

$$x+y+z=6$$

$$2x+3y+z=10$$

$$3x+4y+2z=15$$

Question 2

Find all eigenvalues and corresponding eigenvectors for the matrix A if

$$\begin{bmatrix} 2 & -3 & 0 \\ 2 & -5 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Question 3

Write a Python program (using only **NumPy**) to solve a system of three linear equations in three variables (x,y,z) using the **matrix method**.

- The program should:
 1. Take input of coefficients and constants from the user.
 - A **unique solution**,
 - **No solution**, or
 - **Infinitely many solutions**.
 2. If a unique solution exists, compute it using the matrix method.

Example Input:

$$x+y+z=6$$

$$2x + 3y + z = 10$$

$$3x+4y+2z=15$$

Expected Output:

The program should detect that the system has **no solution**.